12-Week Data Analyst Learning Roadmap

Career goal: Build beginner-level Data Analyst skills and complete a portfolio-ready project. **Profile:** Complete beginner, basic computer knowledge, prefers free resources. **Commitment:** 5–7 hours/week (60–84 hours total).

Weekly plan

Week	Topics & learning objectives	Recommended learning resource	Tools	Practical exercise / completion criterion	Time
1	Spreadsheet fundamentals: rows, columns, data types, sorting, filtering, SUM, AVERAGE, MIN, MAX.	Microsoft Excel Help & Learning: https://support.microsoft.com/en-us/excel/	Google Sheets or Excel	Create a sales sheet with at least 50 records. Use SUM, AVERAGE, MIN and MAX. Save the spreadsheet.	5–7 hrs
2	Spreadsheet analysis: IF, COUNTIF, SUMIF, lookup functions, PivotTables, conditional formatting and charts.	Microsoft Excel Help & Learning: https://support.microsoft.com/en-us/excel/	Google Sheets or Excel	Create a product/region summary table and two charts. Answer five questions about the sales data.	5–7 hrs
3	SQL fundamentals: SELECT, WHERE, ORDER BY, DISTINCT and filtering. Retrieve specific records.	SQLBolt interactive lessons: https://sqlbolt.com/	SQLBolt in browser	Complete introductory exercises and write at least 10 queries that retrieve the correct records.	5–7 hrs
4	SQL analysis: GROUP BY, aggregates, HAVING, JOIN, NULL and subqueries.	SQLBolt lessons: https://sqlbolt.com/	SQLBolt	Solve 10 questions covering total sales, top products, regional performance and a two-table JOIN.	5–7 hrs
5	Python fundamentals: variables, data types, lists, dictionaries, conditions, loops, functions and files.	Kaggle Learn: https://www.kaggle.com/learn/	Kaggle notebooks or Python	Complete introductory exercises and write a program to calculate sales totals and category summaries.	5–7 hrs
6	Pandas: DataFrames, CSV import, selecting, filtering, sorting, grouping and summary statistics.	Pandas tutorials: https://pandas.pydata.org/docs/getting_started/intro_tutorials/	Python, Pandas, Jupyter/Kaggle	Load a CSV, answer 10 questions using Pandas, and export a summary table.	5–7 hrs

12-Week Data Analyst Learning Roadmap — Weekly Plan (continued)

Week	Topics & learning objectives	Recommended learning resource	Tools	Practical exercise / completion criterion	Time
7	Data cleaning: missing values, duplicates, inconsistent text, incorrect data types and outlier investigation.	Pandas tutorials: https://pandas.pydata.org/docs/getting_started/intro_tutorials/	Pandas and spreadsheet software	Clean a dataset with deliberate errors. Report missing values, duplicates, changes and final row count.	5–7 hrs
8	Statistics: mean, median, mode, range, standard deviation, percentiles, distributions and correlation.	Khan Academy statistics: https://www.khanacademy.org/math/probability/probability	Python, Pandas, spreadsheet software	Calculate at least five summary statistics and explain three findings in plain language.	5–7 hrs
9	Visualization: bar/line charts, histograms, scatter plots, labels, legends and avoiding misleading charts.	Matplotlib: https://matplotlib.org/stable/tutorials/index Seaborn: https://github.com/mwaskom/seaborn/blob/master/doc/tutorial/introduction.ipynb	Matplotlib, Seaborn, Pandas	Create at least five readable charts. Each must answer a question and include suitable titles and labels.	5–7 hrs
10	Dashboard development: KPIs, filters, chart layout, interactive reporting and communicating insights.	Microsoft Learn Power BI: https://learn.microsoft.com/training/powerplatform/power-bi/	Power BI Desktop (Windows) or spreadsheet dashboard	Build a dashboard with at least three KPIs, three charts and one useful filter. Validate totals.	5–7 hrs
11	Portfolio analysis: combine cleaning, SQL/Pandas, statistics and visualization; formulate business questions.	SQLBolt: https://sqlbolt.com/ Pandas: https://pandas.pydata.org/docs/getting_started/intro_tutorials/ Kaggle: https://www.kaggle.com/learn/	Python, Pandas, SQL, dashboard tool	Create a first project version with cleaned data, 10 analysis questions, five charts and three evidence-based findings.	5–7 hrs
12	Final portfolio: validation, documentation, project storytelling, limitations, recommendations and reproducibility.	Microsoft Learn Power BI: https://learn.microsoft.com/training/powerplatform/power-bi/ and official docs used earlier.	All selected tools; GitHub or local project folder	Submit README, cleaned data or reproducible cleaning process, code/queries, dashboard, findings, limitations and a 3–5 minute presentation.	5–7 hrs

Free resources and cost notes

- Official documentation, SQLBolt lessons, and open-source Python/Pandas/Matplotlib/Seaborn tools can be used without paying for a course.
- Kaggle Learn offers free learning content; check current account and certificate requirements.
- Excel desktop access may require a licence. Microsoft Learn content is free; Power BI sharing and certification may have separate requirements.
- No free trial is required for this roadmap. Verify current terms before enrolling in any paid course, certificate, or trial.

Final portfolio project: Retail Sales Analysis Dashboard

Analyze sales transactions to identify top products, regional trends, monthly performance, and business opportunities. Deliver: raw and cleaned data (or reproducible cleaning process), Python notebook/script, SQL queries, five or more charts, dashboard/report, README, findings, limitations, and a short presentation.

Skills to demonstrate after 12 weeks

- Spreadsheets: formulas, PivotTables and summaries; SQL: filtering, aggregation and joins.
- Python fundamentals; Pandas data import, cleaning, transformation and analysis.
- Descriptive statistics and correlation; appropriate charts and dashboard design.
- Data validation, evidence-based communication, and a documented, reproducible portfolio project.